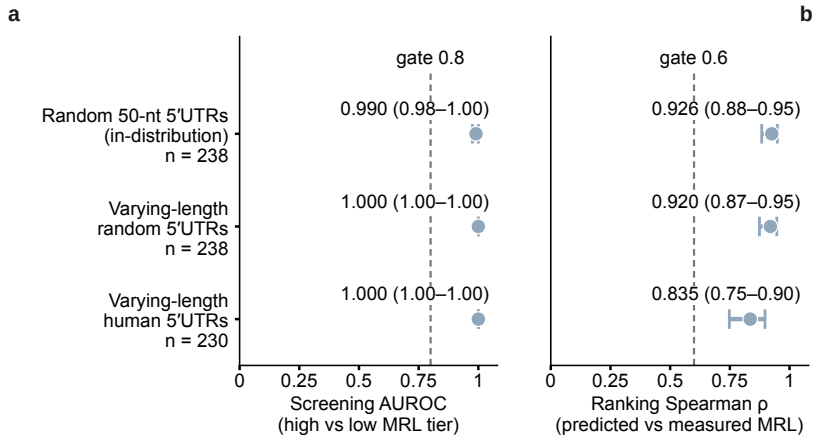
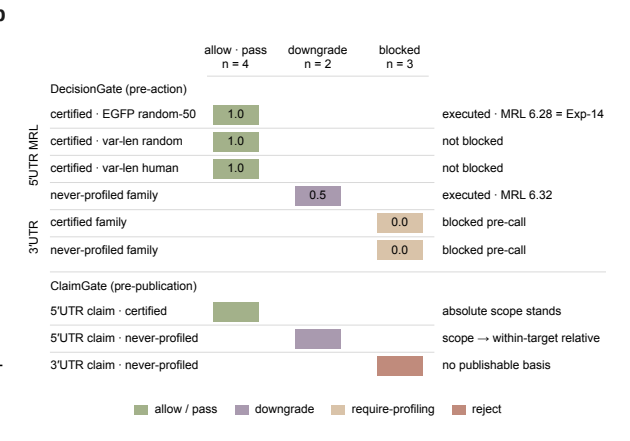
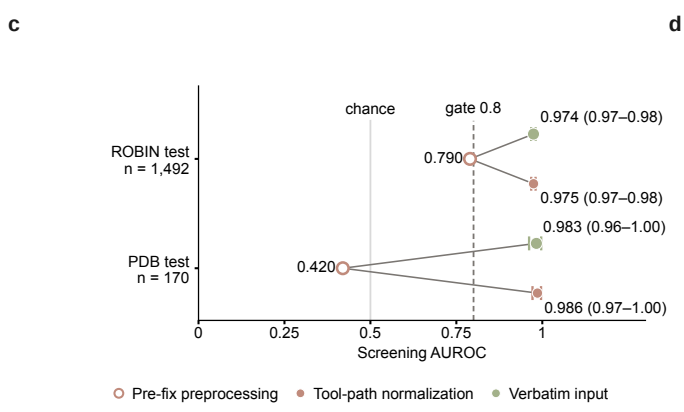
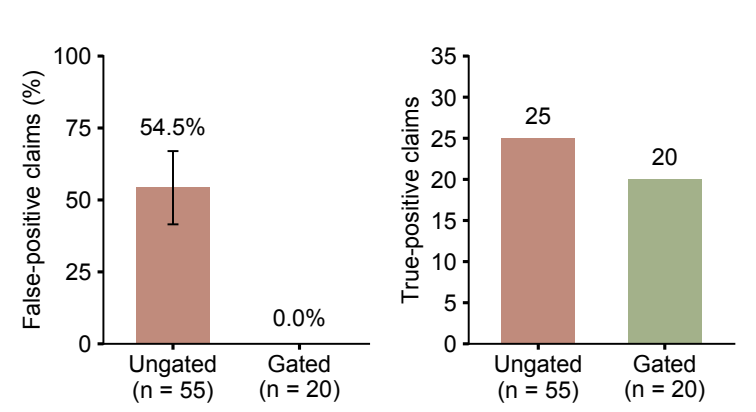
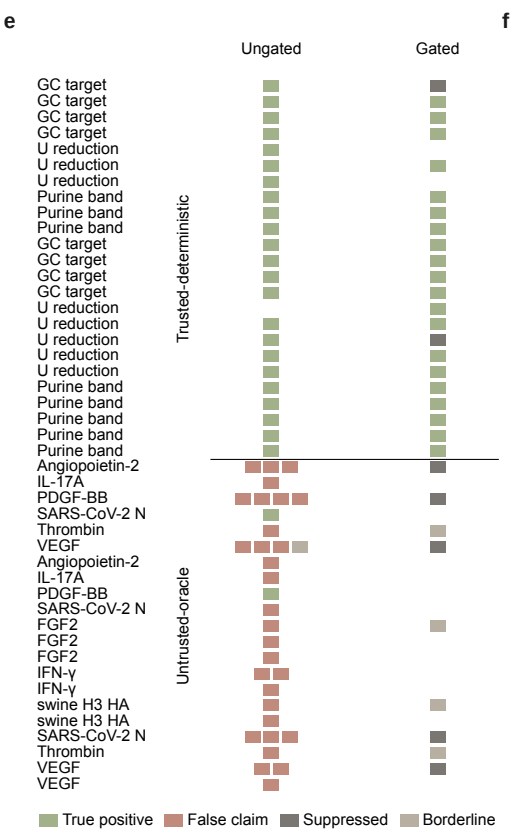
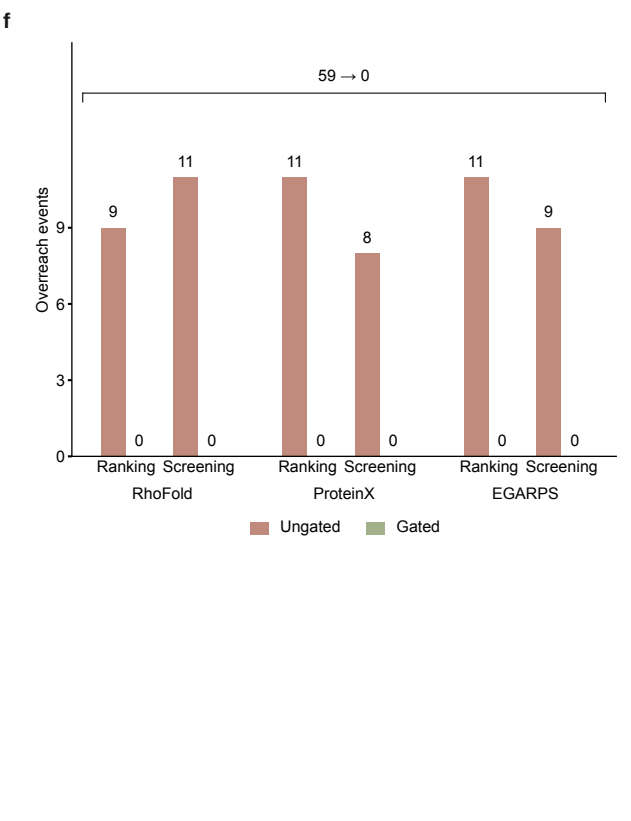

a
gate 0.8
gate 0.6
Random 50-nt 5′UTRs (in-distribution) n = 238
Varying-length random 5′UTRs n = 238
Varying-length human 5′UTRs n = 230
0.990 (0.98–1.00)
1.000 (1.00–1.00)
1.000 (1.00–1.00)
0.926 (0.88–0.95)
0.920 (0.87–0.95)
0.835 (0.75–0.90)
0 0.25 0.5 0.75 1
Screening AUROC (high vs low MRL tier)
Ranking Spearman ρ (predicted vs measured MRL)

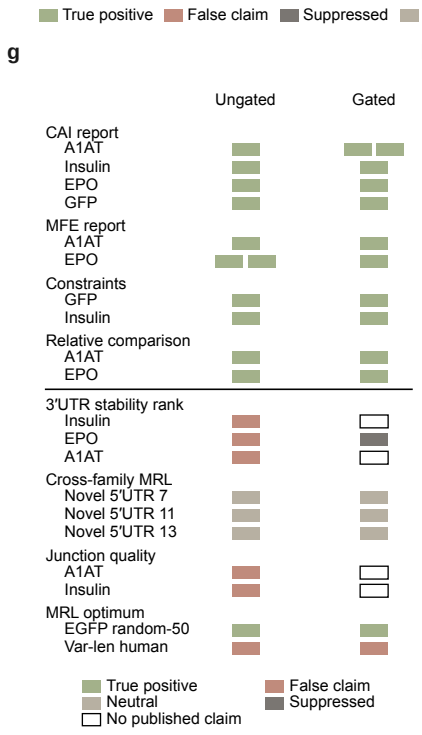

b
allow · pass n = 4
downgrade n = 2
blocked n = 3
DecisionGate (pre-action)
5′UTR MRL
3′UTR
certified · EGFP random-50
certified · var-len random
certified · var-len human
never-profiled family
certified family
never-profiled family
executed · MRL 6.28 = Exp-14
not blocked
not blocked
executed · MRL 6.32
blocked pre-call
blocked pre-call
ClaimGate (pre-publication)
5′UTR claim · certified
5′UTR claim · never-profiled
3′UTR claim · never-profiled
absolute scope stands
scope → within-target relative
no publishable basis
allow / pass
downgrade
require-profiling
reject

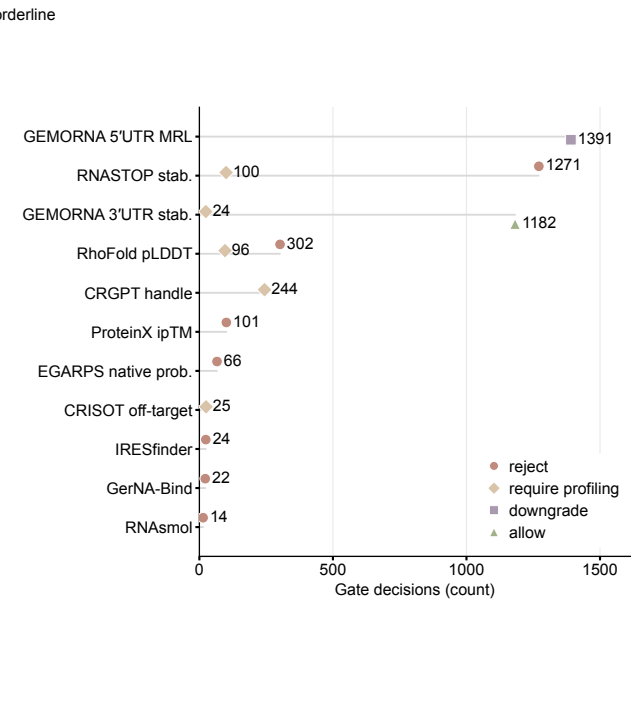

c
chance
gate 0.8
ROBIN test n = 1,492
PDB test n = 170
0.790
0.420
0.974 (0.97–0.98)
0.975 (0.97–0.98)
0.983 (0.96–1.00)
0.986 (0.97–1.00)
Screening AUROC
Pre-fix preprocessing
Tool-path normalization
Verbatim input

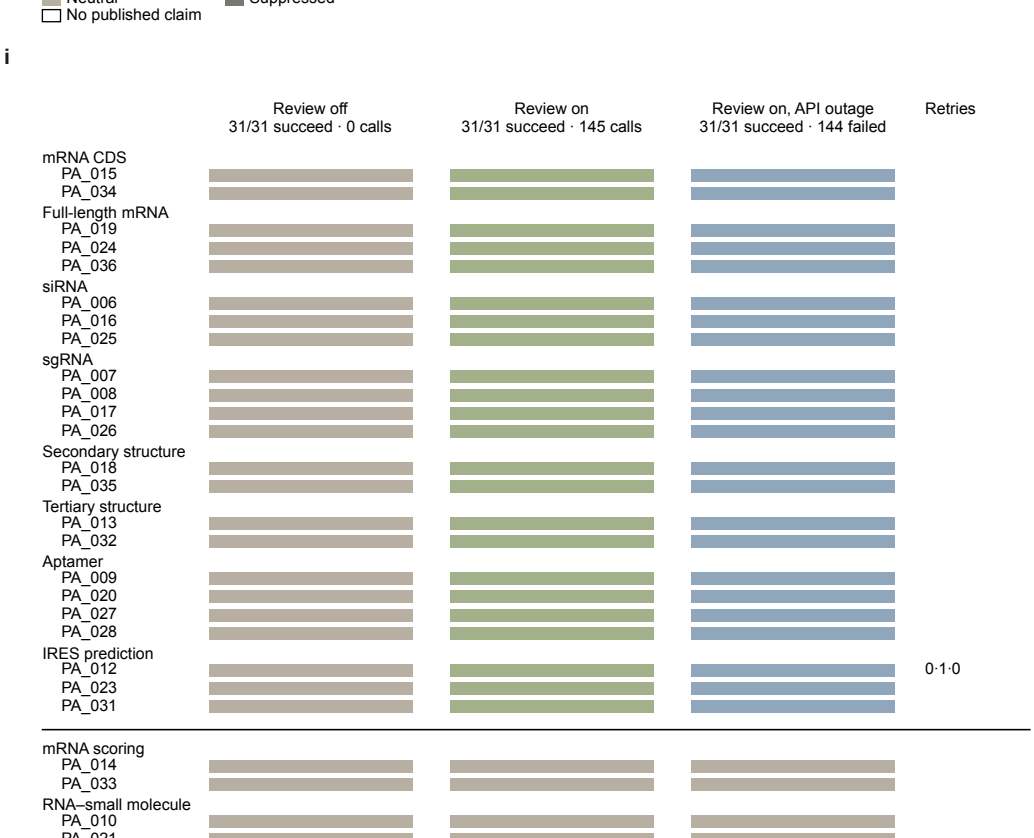

d
False-positive claims (%)
54.5%
0.0%
Ungated (n = 55)
Gated (n = 20)
True-positive claims
25
20

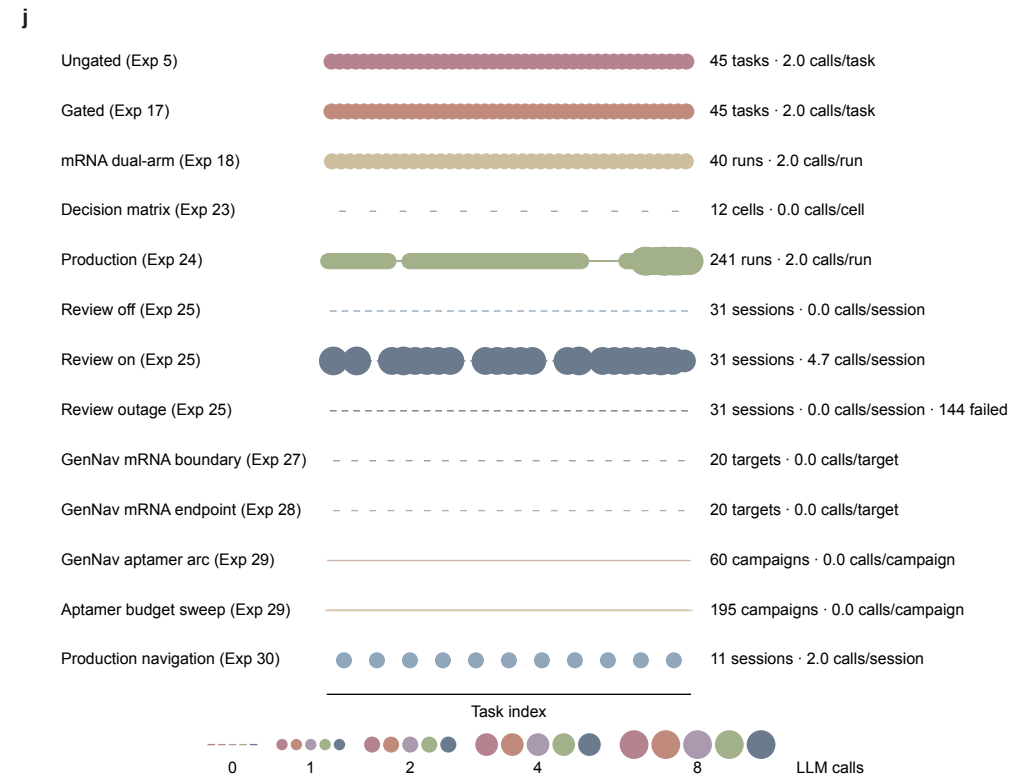

e
Ungated
Gated
Trusted-deterministic
Untrusted-oracle
GC target
U reduction
Purine band
Angiopoietin-2
IL-17A
PDGF-BB
SARS-CoV-2 N
Thrombin
VEGF
FGF2
IFN-γ
swine H3 HA
True positive
False claim
Suppressed
Borderline


f
59 → 0
Overreach events
RhoFold
ProteinX
EGARPS
Ranking
Screening
Ungated
Gated


g
Ungated
Gated
CAI report
MFE report
Constraints
Relative comparison
3′UTR stability rank
Cross-family MRL
Junction quality
MRL optimum
A1AT
Insulin
EPO
GFP
Novel 5′UTR 7
Novel 5′UTR 11
Novel 5′UTR 13
EGFP random-50
Var-len human
True positive
False claim
Neutral
Suppressed
No published claim


h
GEMORNA 5′UTR MRL
RNASTOP stab.
GEMORNA 3′UTR stab.
RhoFold pLDDT
CRGPT handle
ProteinX ipTM
EGARPS native prob.
CRISOT off-target
IRESfinder
GerNA-Bind
RNAsmol
1391
1271
100
24
1182
96
302
244
101
66
25
24
22
14
reject
require profiling
downgrade
allow
Gate decisions (count)


i
Review off 31/31 succeed · 0 calls
Review on 31/31 succeed · 145 calls
Review on, API outage 31/31 succeed · 144 failed
Retries
mRNA CDS
Full-length mRNA
siRNA
sgRNA
Secondary structure
Tertiary structure
Aptamer
IRES prediction
mRNA scoring
RNA–small molecule
ASO screening
PA_015
PA_034
PA_019
PA_024
PA_036
PA_006
PA_016
PA_025
PA_007
PA_008
PA_017
PA_026
PA_018
PA_035
PA_013
PA_032
PA_009
PA_020
PA_027
PA_028
PA_012
PA_023
PA_031
0·1·0
PA_014
PA_033
PA_010
PA_021
PA_029
PA_011
PA_022
PA_030
Succeeds · review completed
Succeeds · review unavailable
Succeeds · no review


j
Ungated (Exp 5)
Gated (Exp 17)
mRNA dual-arm (Exp 18)
Decision matrix (Exp 23)
Production (Exp 24)
Review off (Exp 25)
Review on (Exp 25)
Review outage (Exp 25)
GenNav mRNA boundary (Exp 27)
GenNav mRNA endpoint (Exp 28)
GenNav aptamer arc (Exp 29)
Aptamer budget sweep (Exp 29)
Production navigation (Exp 30)
45 tasks · 2.0 calls/task
45 tasks · 2.0 calls/task
40 runs · 2.0 calls/run
12 cells · 0.0 calls/cell
241 runs · 2.0 calls/run
31 sessions · 0.0 calls/session
31 sessions · 4.7 calls/session
31 sessions · 0.0 calls/session · 144 failed
20 targets · 0.0 calls/target
20 targets · 0.0 calls/target
60 campaigns · 0.0 calls/campaign
195 campaigns · 0.0 calls/campaign
11 sessions · 2.0 calls/session
Task index
LLM calls